THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A body moving with uniform velocity has acceleration: 1

(a) 10 m/s squared **(b)** infinite **(c)** zero **(d)** 1 m/s squared

2. Weight = 1

(a) mg **(b)** m + g **(c)** m/g **(d)** g/m

3. Rough surface has: 1

(a) no mass **(b)** more friction **(c)** less friction **(d)** no friction

4. The force of gravity acts: 1

(a) away from the Earth **(b)** towards the centre of the Earth

(c) sideways **(d)** randomly

5. g on Earth is about: 1

(a) 0 **(b)** 10 m/s squared **(c)** 100 m/s squared **(d)** 1 m/s squared

6. The mass of a body on the Moon: 1

(a) becomes one-sixth **(b)** does not change

(c) doubles **(d)** becomes zero

7. The brakes of a vehicle work due to: 1

(a) friction **(b)** inertia only **(c)** magnetism **(d)** gravity

8. Friction acts: 1

(a) never **(b)** upward

(c) along motion **(d)** opposite to motion

9. The weight on the Moon is: 1

(a) same **(b)** one-sixth of that on Earth

(c) zero **(d)** six times

10. Seat belts protect us due to: 1

(a) inertia **(b)** magnetism **(c)** friction **(d)** gravity

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the weight of a 6 kg body (g = 10). 2

12. What is inertia? 2

13. What is weight? State its formula. 2

14. Name two uses of friction. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. State three uses of friction. 3

16. Why are seat belts used? State three points. 3

17. State the effect of force on a body: any three. 3

OR

Find the acceleration of a 10 kg body with a force of 50 N. 3

18. Explain why a rolling ball stops after some time. 3

OR

Why does a heavy object need a greater force to start moving? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain friction, its advantages and its disadvantages. 5

OR

Describe an activity to show inertia of rest. 5

20. Explain mass and weight with a numerical example. 5

OR

State the effects of friction in daily life with five examples. 5

END OF QUESTION PAPER Total: 40 Marks

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